

Personalized Recommendation System

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Model Architectures

0.9321

Best RMSE (Hybrid)

0.0085

Best MAP@10 (NCF)

Problem Overview

SLIDE 02

⚡ Extreme Sparsity

Only 1.2% of user-item pairs observed. 99.38% of the interaction matrix is empty — models must generalise from sparse signal.

🎯 Two Conflicting Objectives

RMSE and MAP@10 often pull in opposite directions. No single model dominates both dimensions simultaneously.

🐘 Scale & Cold Start

100M ratings, 480K users, 17.7K movies. New users with no history require dedicated fallback strategies.

100M+

Total Ratings

480K

Unique Users

17,770

Movies

4.8M

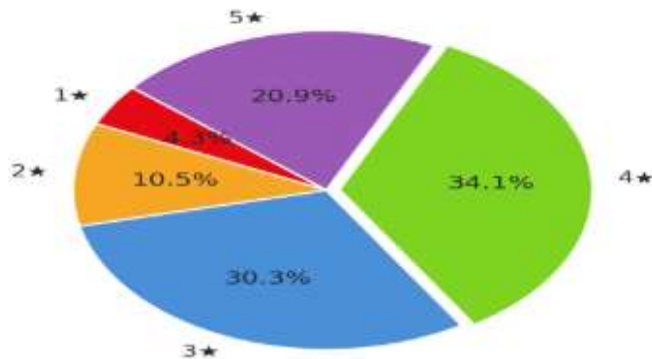
Ratings (our subset)

Task: Learn user preferences → Predict unseen ratings → Generate Top-K personalised recommendations

Exploratory Data Analysis

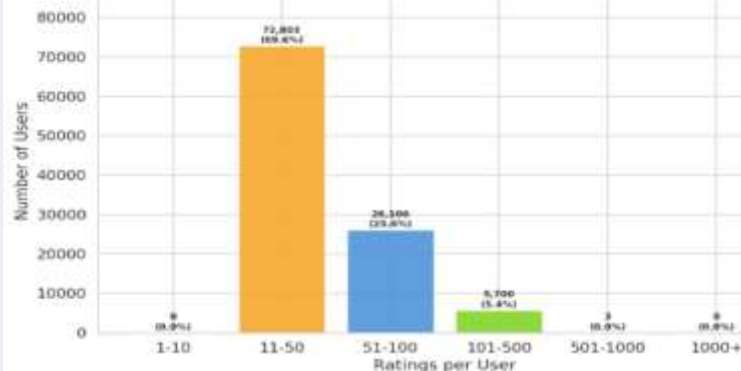
SLIDE 03

Rating Share (%)



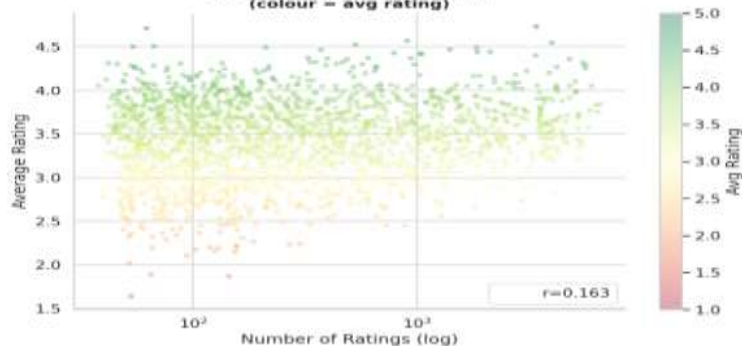
Modal rating = 4★ · Selection bias observed

User Activity Segments



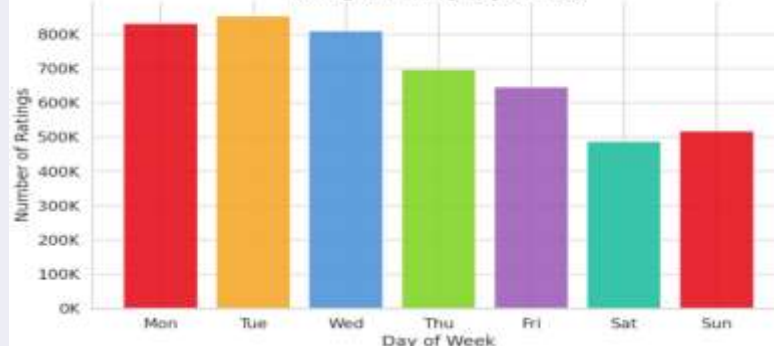
Top 5% users → 40%+ of all ratings · Power-law

Popularity vs Average Rating
(colour = avg rating)



Popularity bias risk · Long-tail distribution

Rating Volume by Day of Week



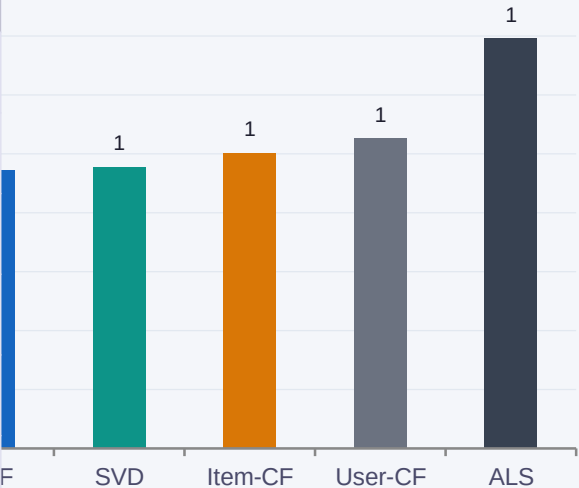
1999–2005 growth · Temporal drift noted

Model Comparison & Results

SLIDE 04

Model	RMSE ↓	MAE ↓	R ²	MAP@10 ↑	NDCG@10 ↑	Coverage
Hybrid ★	0.9321	0.7345	0.2025	0.0040	0.0079	19.1%
NCF	0.9426	0.7374	0.1845	0.0085	0.0159	2.2%
SVD	0.9566	0.7445	0.1601	0.0028	0.0062	18.4%
Item-CF	1.0053	0.7715	0.0724	0.0013	0.0034	19.7%
User-CF	1.0514	0.8841	-0.015	—	—	—
ALS	1.3928	1.0779	-0.781	—	—	—

RMSE by Model (lower = better)



Best RMSE



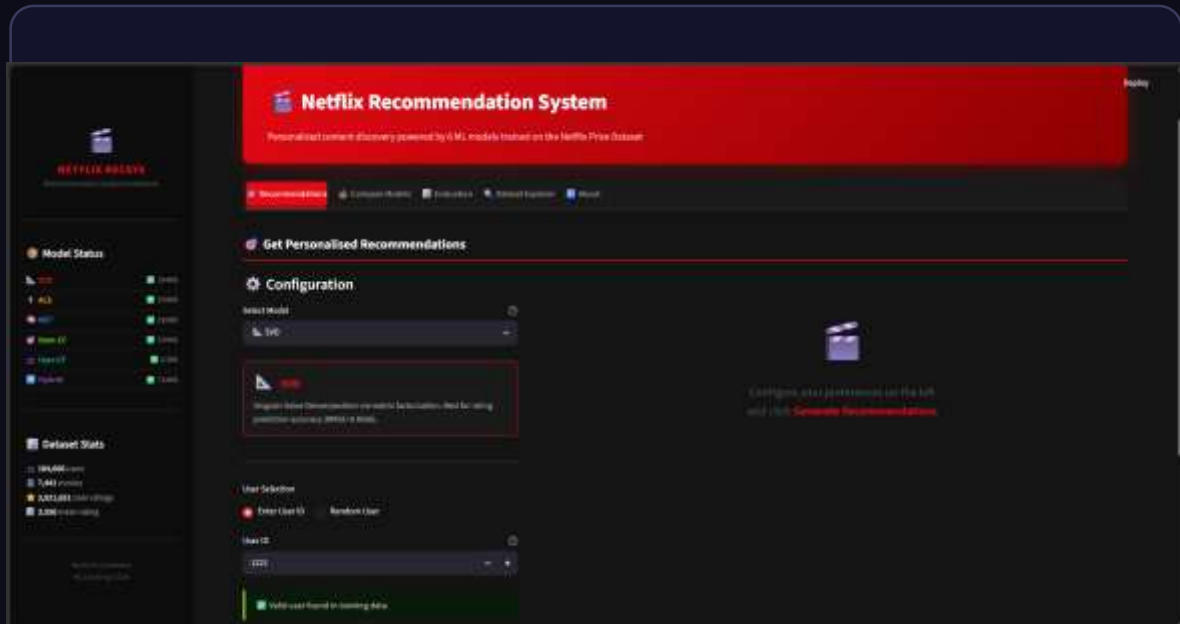
Best MAP@10

★ Hybrid = NCF (63.8%) + SVD (36.2%) · MAP@10 threshold ≥ 3.5★

Interactive Dashboard

SLIDE 05

Streamlit App · `app/streamlit_app.py`



Top-10 Recs

Per-user recommendations with predicted ratings

Model Selector

Switch between all 6 models live

Similar Content

Item-similarity exploration for any title

Recommendation Examples

SLIDE 06

✓ SUCCESS CASE — Hybrid Model

User	User #8241
History	Schindler's List (5★) · Shawshank Redemption (5★) Se7en (4★) · American History X (4★)
Top-3	1. The Green Mile (1999) 2. Fight Club (1999) 3. Memento (2000)
Why	Hybrid correctly maps user to intense drama/thriller latent region. Strong qualitative match via NCF non-linear layers.

⚠ COLD-START CASE — Fallback

User	User #99871 (new user, 2 ratings)
History	Titanic (4★) · The Dark Knight (5★)
Top-3	1. The Dark Knight Rises (2012) 2. Batman Begins (2005) 3. Inception (2010)
Why	< 2 ratings → latent models insufficient. Fallback: item-item similarity on Dark Knight + global popularity. Reasonable but limited.

Key Insights

SLIDE 07

01 Coverage ↔ Accuracy Tradeoff is Real

NCF: MAP@10 0.0085 but only 2.2% catalogue coverage.

Item-CF: 19.7% coverage but lower ranking precision.

No single model dominates — production systems must choose an operating point based on business objective.

02 Ensemble Works Because Models Differ

SVD ↔ NCF recommendation Jaccard overlap: only 12.2%.

Ridge stacking learns: NCF 63.8% + SVD 36.2%.

Diversity between base models is what makes ensembling effective — not just combining two good models.

03 NCF's Ranking Edge Comes at a Cost

Non-linear GMF+MLP layers give NCF the best MAP@10 of all individual models.

But it learns to recommend a small 'safe' set — effectively a popularity bias in ranking space.

Mitigation: re-ranking with diversity injection (MMR, DPP).

04 Low MAP@10 is an Evaluation Artefact

Test set only contains movies users actually rated — unobserved but relevant movies get no credit.

This is the missing data problem in IR evaluation — consistent with literature on explicit feedback benchmarks.

Values of 0.003–0.009 are expected and not a model failure.

6

Models Built

0.9321

Best RMSE

12.2%

SVD $\leftrightarrow$ NCF Jaccard Overlap

19.7%

Max Catalogue Coverage

Building the right recommender
is not about the best model —
it's about the right tradeoff.

 github.com/25f1001872/netflix-recsys

Future: Temporal SVD · Content metadata · LightGCN · MMR re-ranking · FastAPI deployment